OXFORD AQA

INTERNATIONAL QUALIFICATIONS

Please write clearly in block capitals.

Centre number | | | | | |

Candidate number | | | | |

Surname ____________________

Forename(s) ____________________

Candidate signature ____________________

I declare this is my own work.

INTERNATIONAL GCSE
PHYSICS

Paper 2

Wednesday 13 November 2024 07:00 GMT Time allowed: 1 hour 30 minutes

Materials

For this paper you must have:
- a pencil and a ruler
- a scientific calculator
- a protractor
- the Physics Equations Sheet (enclosed).

Instructions
- Use black ink or black ball-point pen.
- Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions in the spaces provided.
- If you need extra space for your answer(s), use the lined pages at the end of this book. Write the question number against your answer(s).
- Do all rough work in this book. Cross through any work you do not want to be marked.
- In all calculations, show clearly how you worked out your answer.

Information
- The maximum mark for this paper is 90.
- The marks for questions are shown in brackets.
- You are expected to use a calculator where appropriate.

For Examiner's Use	
Question	Mark
1	
2	
3	
4	
5	
6	
7	
8	
TOTAL	

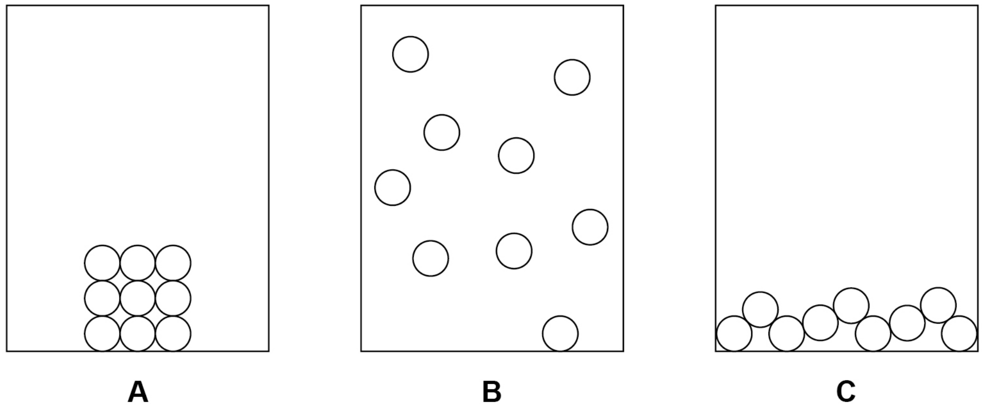
Do not write outside the box

Answer **all** questions in the spaces provided.

0 1 An insulated cup is designed to keep a liquid hot.

Figure 1 shows the arrangement of particles in the three states of matter.

Figure 1



0 1 . 1 Which diagram shows the arrangement of particles in a liquid?

[1 mark]

Tick (✓) **one** box.

A ☐

B ☐

C ☐

0 1 . 2 Which **one** of the following describes the movement of particles in a liquid?

[1 mark]

Tick (✓) **one** box.

Particles are in a fixed position.	☐
Particles move in different directions.	☐
Particles move in the same direction.	☐

Question 1 continues on the next page

Do not write outside the box

Figure 2 shows an insulated cup.

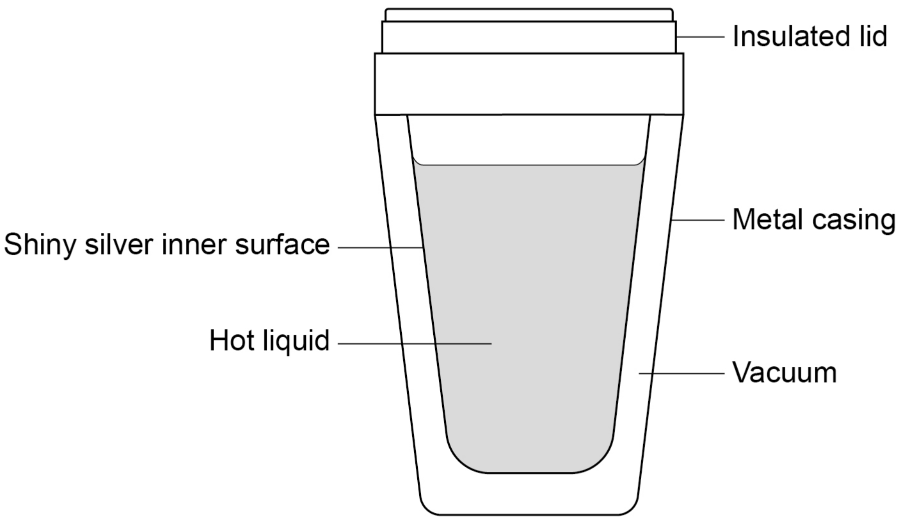
The insulated cup has a metal casing which contains a vacuum.

The vacuum contains very few particles.

The inner surface of the metal casing is coloured shiny silver.

Figure 2 shows a cross-section of the insulated cup.

Figure 2



0 1 . 3 Give **one** type of energy transfer that is reduced by the shiny silver inner surface shown in **Figure 2**.

[1 mark]

Tick (✓) **one** box.

Conduction ☐

Convection ☐

Evaporation ☐

Radiation ☐

Do not write outside the box

0 1 . 4 A vacuum contains very few particles.

Which **two** types of energy transfer will be reduced by the vacuum in the metal casing shown in **Figure 2**?

[2 marks]

Tick (✓) **two** boxes.

Conduction ☐

Convection ☐

Evaporation ☐

Radiation ☐

The insulated cup has a metal casing which is a good thermal conductor.

0 1 . 5 Explain why metals are good thermal conductors.

[2 marks]

____

____

____

____

0 1 . 6 Suggest **one** reason why a metal casing is used even though metals are good thermal conductors.

[1 mark]

____

____

8

0 2 **Table 1** shows the range of frequencies of sound waves that different animals can hear.

Table 1

Animal	**Lowest frequency in hertz**	**Highest frequency in hertz**
Bat	20	160 000
Chicken	125	2 000
Whale	800	13 500

0 2 . 1 Which animal in **Table 1** can hear the highest frequency sound wave?

[1 mark]

____

0 2 . 2 Give the **lowest** frequency of sound wave that can be heard by **both** chickens and whales.

[1 mark]

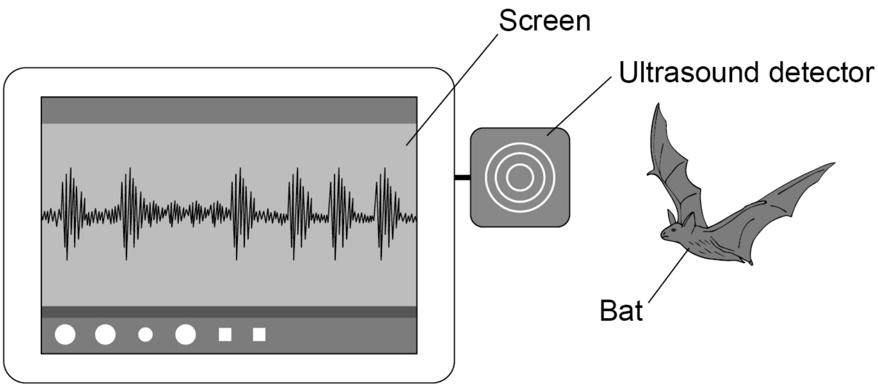
Lowest frequency = ______________________________ Hz

Do not write outside the box

A bat is a small animal that flies at night. The bat emits ultrasound waves as it flies.

Figure 3 shows an ultrasound detector connected to a screen.

Figure 3



0 2 . 3 What type of wave is an ultrasound wave?

[1 mark]

Tick (✓) **one** box.

Electromagnetic ☐

Longitudinal ☐

Transverse ☐

Question 2 continues on the next page

Do not write outside the box

0 2 . 4 The bat emits an ultrasound wave with a frequency of 60 000 Hz.

Calculate the wavelength of the ultrasound wave.

speed of ultrasound wave = 330 m/s

Use the Physics Equations Sheet.

[3 marks]

Wavelength = ____________________ m

0 2 . 5 A human needs a screen attached to the ultrasound detector to know that the bat was emitting an ultrasound wave.

Explain why.

[2 marks]

0 2 . 6 Ultrasound waves have medical uses.

Give **one** medical use of ultrasound waves.

[1 mark]

9

Do not write outside the box

Turn over for the next question

DO NOT WRITE ON THIS PAGE
ANSWER IN THE SPACES PROVIDED

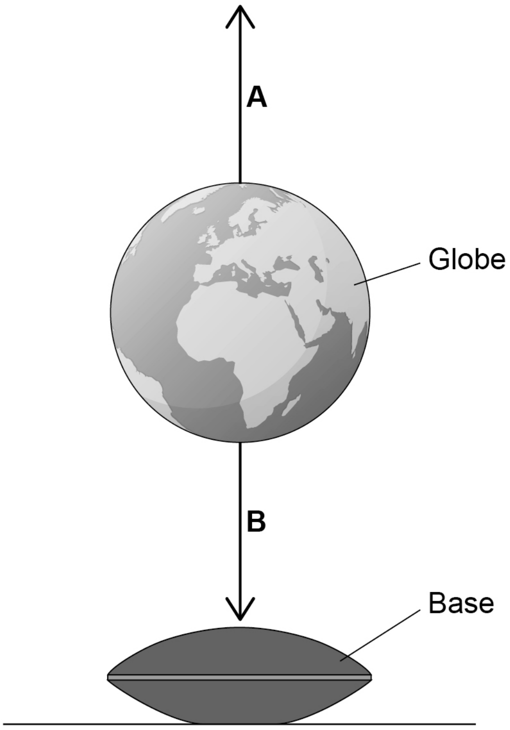
Do not write outside the box

0 3 **Figure 4** shows a globe which is stationary in the air above a base.

The globe contains a permanent magnet.

The base contains an electromagnet.

Figure 4



0 3 . 1 The forces acting on the globe are represented by arrows in **Figure 4**.

Name force **A** and force **B**.

Choose answers from the box.

[2 marks]

air resistance	**electrostatic**	**friction**	**magnetic**	**weight**

Force **A** ____________________________

Force **B** ____________________________

Do not write outside the box

The resultant force acting on the globe is zero.

0 3 . 2 Explain how the arrows in **Figure 4** show that the resultant force acting on the globe is zero.

[2 marks]

____

____

____

____

0 3 . 3 The globe in **Figure 4** has a mass of 0.20 kg.

gravitational field strength = 9.8 N/kg

Calculate the downward force acting on the globe in **Figure 4**.

Use the equation:

$$\text{force} = \text{mass} \times \text{gravitational field strength}$$

[2 marks]

____

____

____

Force = ____________________ N

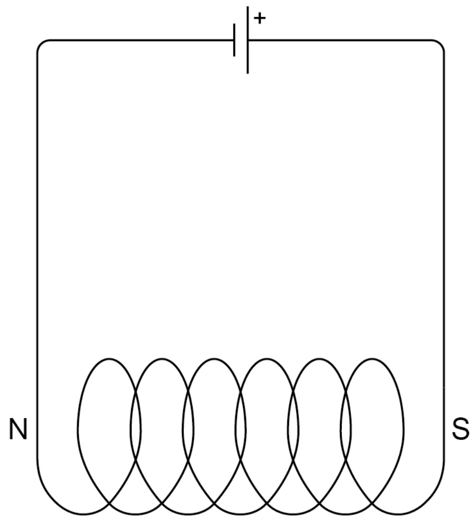
Question 3 continues on the next page

Do not write outside the box

The electromagnet in the base is made using a solenoid with an iron core.

Figure 5 shows a solenoid with the north pole (N) and south pole (S) labelled.

Figure 5



0 3 . 4 Draw the magnetic field pattern around the solenoid in **Figure 5**.

[2 marks]

0 3 . 5 How would reversing the direction of the current in the solenoid affect the magnetic field around the solenoid?

[1 mark]

____

____

0 3 . 6 How does adding an iron core to the solenoid affect the magnetic field around the solenoid?

[1 mark]

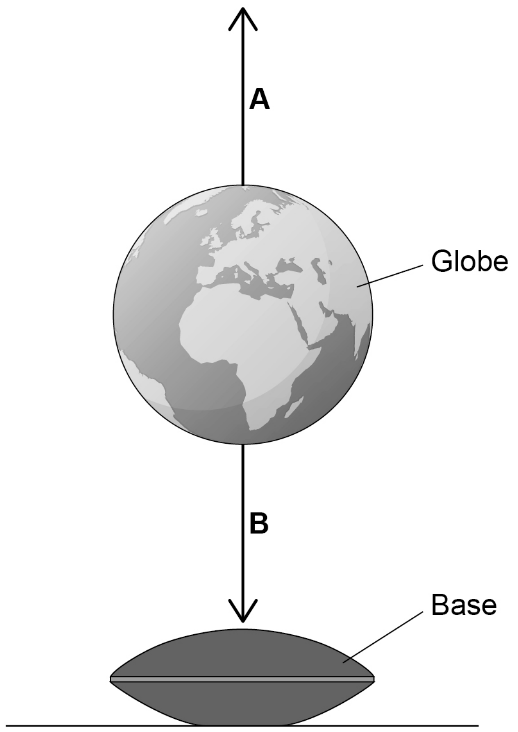
____

____

Do not write outside the box

Figure 4 is repeated below.

Figure 4



0 3 . 7 Explain what would happen to the globe if the current in the electromagnet was decreased.

[3 marks]

13

Turn over for the next question

Do not write outside the box

0 4 A student investigated how the current in a resistor changed as the potential difference across the resistor was changed.

The resistor was kept at a constant temperature.

The student used the following equipment:

- battery
- resistor
- switch
- variable resistor
- ammeter
- voltmeter
- connecting wires.

0 4 . 1 Describe a method the student could use.

Include a circuit diagram.

[6 marks]

Do not write outside the box

Extra space

0 4 . 2 Why should the current in the resistor be kept small during the investigation?

[1 mark]

Question 4 continues on the next page

Table 2 shows some of the results.

Table 2

Potential difference in volts	Current in amps
0.0	0.05
1.0	0.12
2.0	0.20
3.0	0.27
4.0	0.33
5.0	0.40
6.0	0.46
7.0	0.53
8.0	0.60

Do not write outside the box

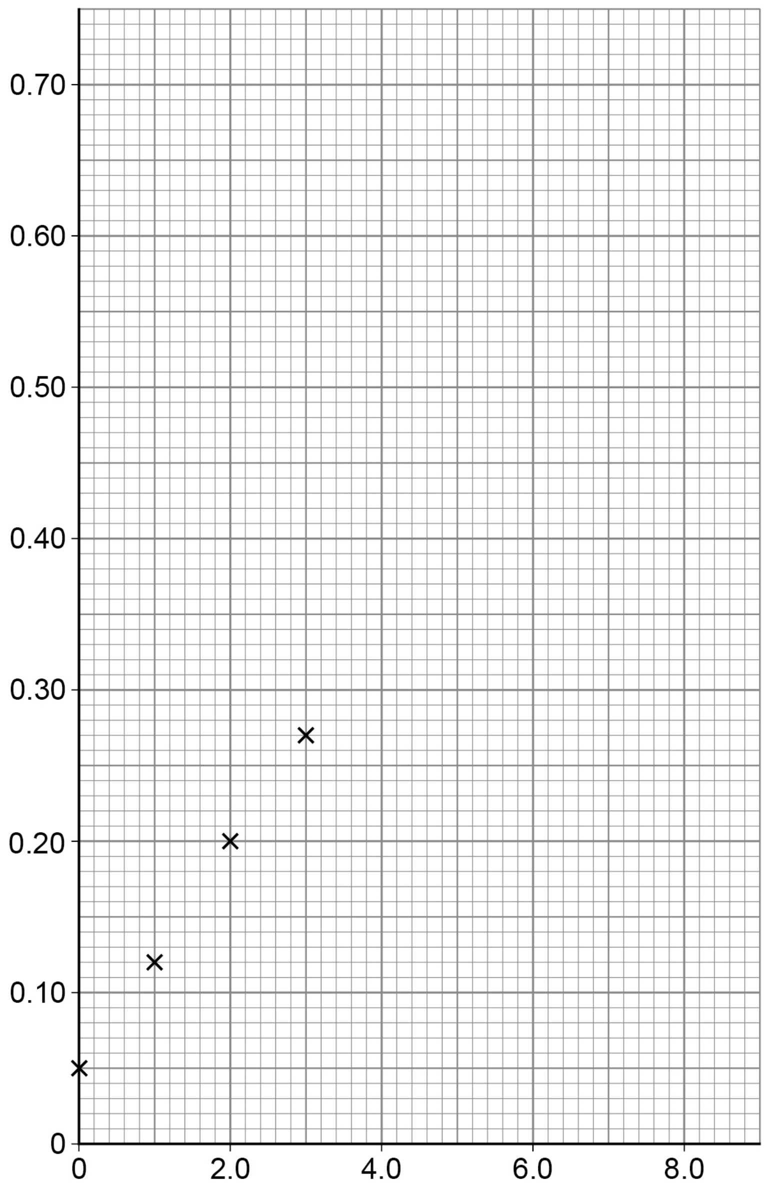
0 4 . 3 Complete **Figure 6**.

You should:

- label the axes
- plot the remaining results from **Table 2**
- draw a line of best fit.

[4 marks]

Figure 6



Question 4 continues on the next page

Do not write outside the box

0 4 . 4 Explain how the graph shows there is a systematic error in the results.

[3 marks]

14

Do not write outside the box

0 5 Speed affects the braking distance of a vehicle.

0 5 . 1 Give **two** other factors that affect the braking distance of a vehicle.

[2 marks]

Tick (✓) **two** boxes.

Age of the driver ☐

Condition of the vehicle's brakes ☐

Driver drinking alcohol ☐

Mass of the vehicle ☐

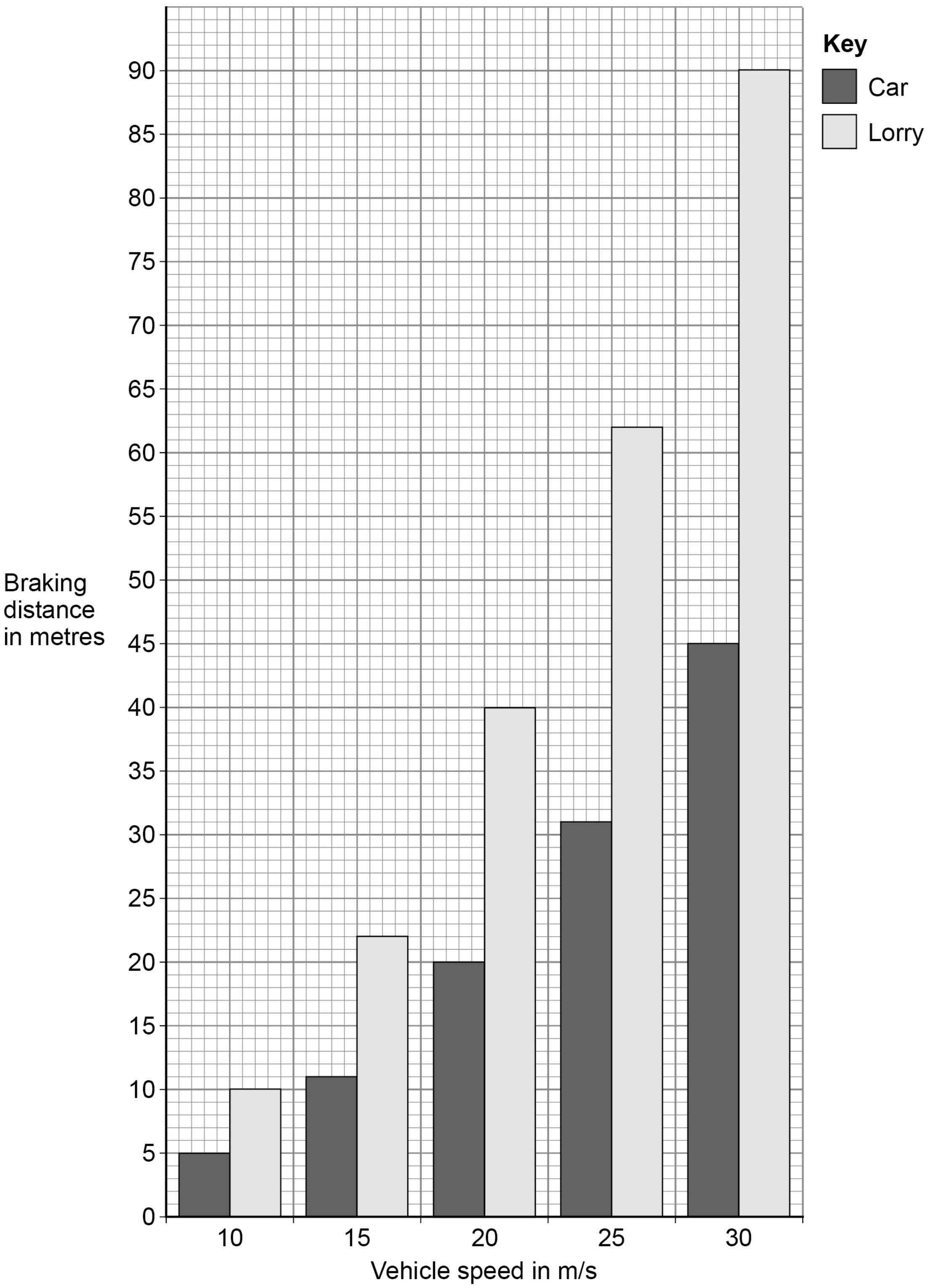
Tiredness of the driver ☐

Question 5 continues on the next page

Figure 7 shows the braking distance for a car and the braking distance for a lorry.

The braking distances were measured in the same conditions.

Figure 7



Do not write outside the box

0 5 . 2 What happens to the braking distance as the speed increases?

Use **Figure 7**.

[1 mark]

0 5 . 3 Compare the braking distance of the car and the braking distance of the lorry at 20 m/s.

Use data from **Figure 7**.

[2 marks]

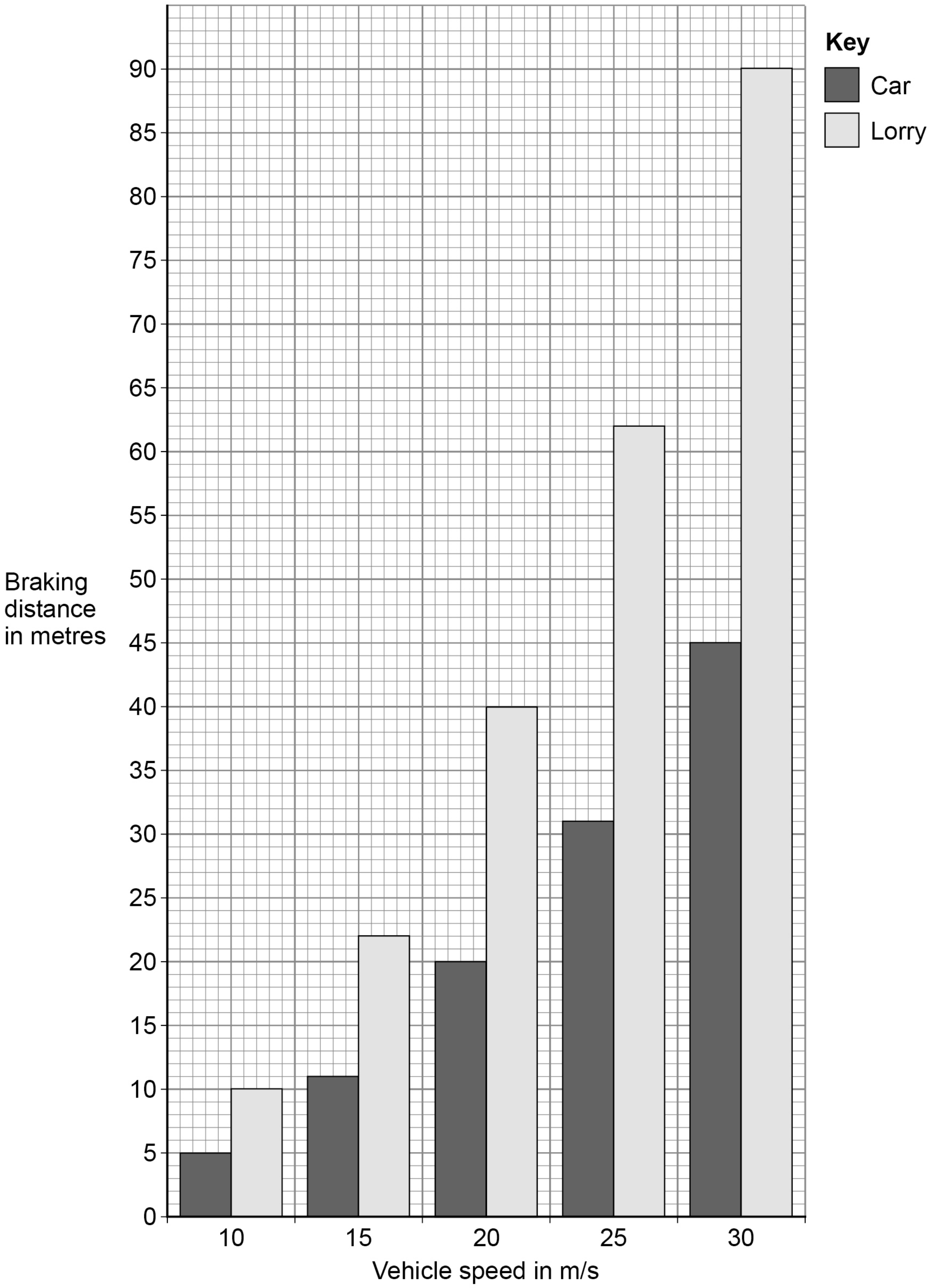
0 5 . 4 Give the reason why stopping distance is greater than braking distance.

[1 mark]

Question 5 continues on the next page

0 5 . 5 **Figure 7** is repeated below.

Figure 7



The car in **Figure 7** is travelling at 30 m/s.

A constant braking force is applied by the brakes until the car stops.

The work done by the brakes is 2.7 MJ.

Determine the braking force.

Use **Figure 7**.

[4 marks]

Braking force = ______________________ N

10

Turn over for the next question

Do not write outside the box

0 6 Solar panels are used to generate electricity.

0 6 . 1 Give **two** factors that could affect the power output of solar panels.

[2 marks]

1 ____

____

2 ____

____

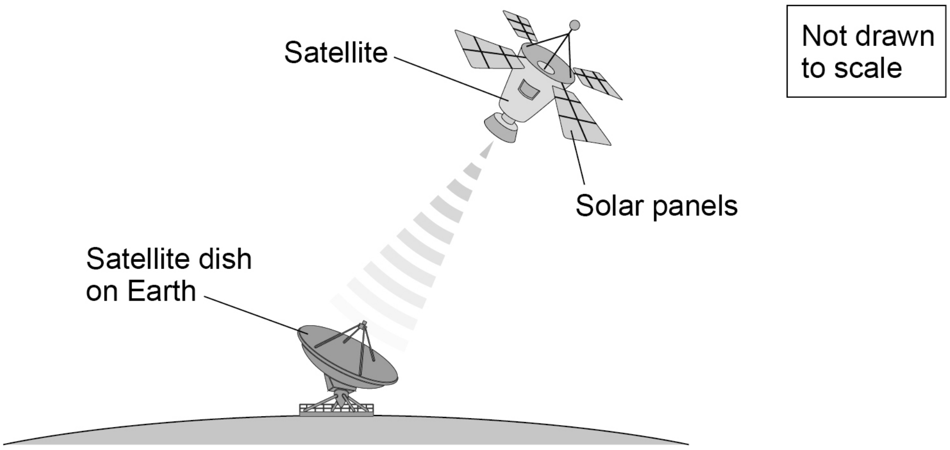
Scientists are trying to develop a space-based solar power station using satellites.

The electricity generated is converted into radio waves which are transmitted to Earth.

A satellite dish on Earth receives the radio waves.

Figure 8 shows a model of a space-based solar power station using a satellite.

Figure 8



Do not write outside the box

0 6 . 2 The model had a power input of 30 W and produced a useful power output of 7.5 W.

Calculate the efficiency of the system.

Use the Physics Equations Sheet.

[2 marks]

Efficiency = ____________

0 6 . 3 Suggest **one** advantage and **one** disadvantage of using solar panels in space compared to using solar panels on Earth.

[2 marks]

Advantage ____________

Disadvantage ____________

Question 6 continues on the next page

Do not write outside the box

0 6 . 4 The satellite will orbit the Earth in a circle.

The speed of the satellite in its orbit will be 3100 m/s.

The time taken to complete one orbit will be 24 hours.

Calculate the radius of the orbit.

Give your answer to 2 significant figures.

circumference of a circle = $2\pi r$

$\pi = 3.14$

[5 marks]

Radius of orbit (2 significant figures) = ____________________ m

Do not write outside the box

0 6 . 5 A different satellite has a greater orbital radius than the satellite in Question **06.4**.

Explain how the speed of this satellite is different.

[2 marks]

0 6 . 6 Different countries are working together to develop a space-based solar power station.

Suggest why different countries working together is an advantage.

[1 mark]

14

Turn over for the next question

Do not write outside the box

There are no questions printed on this page

DO NOT WRITE ON THIS PAGE
ANSWER IN THE SPACES PROVIDED

Do not write outside the box

0 7 **Figure 9** shows a child on a swing.

The child oscillates backwards and forwards on the swing.

Figure 9

0 7 . 1 The period of the oscillation is 2.5 s.

Calculate the frequency of the oscillation.

Use the equation:

$$\text{period} = \frac{1}{\text{frequency}}$$

[3 marks]

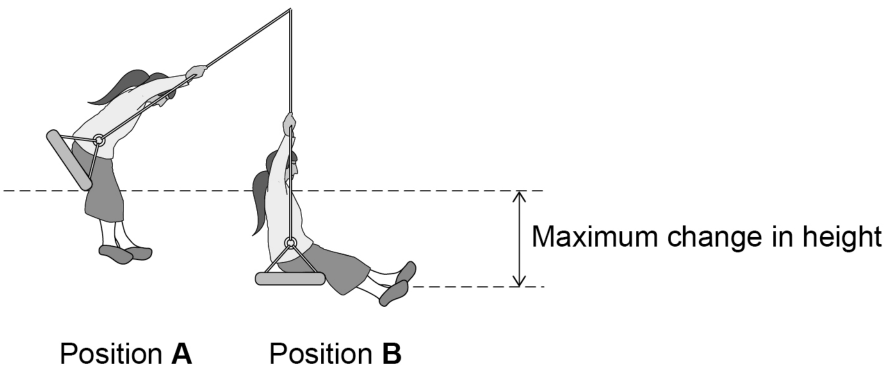
Frequency = ____________________ Hz

Question 7 continues on the next page

0 7 . 2 **Figure 10** shows different positions of the child and swing whilst oscillating.

In Position **A** the child is at the maximum height. In Position **B** the child is at the minimum height.

Figure 10



The maximum change in height shown in **Figure 10** is 62.5 cm.

Calculate the maximum speed of the child and swing.

mass of child and swing = 40 kg

gravitational field strength = 9.8 N/kg

[6 marks]

Maximum speed = ____________________ m/s

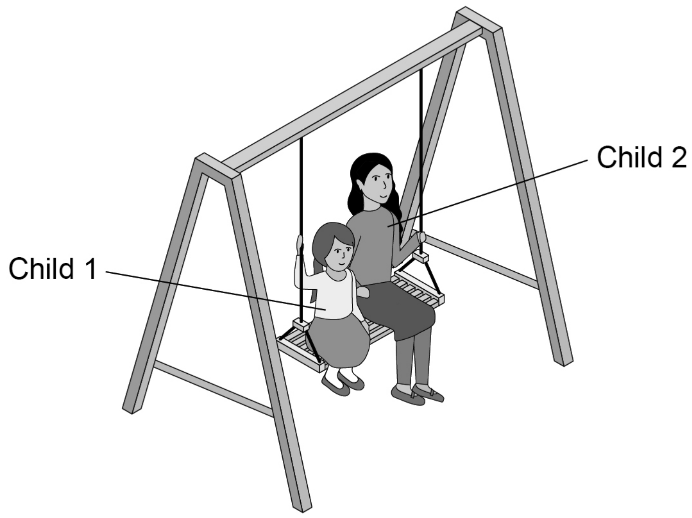
Do not write outside the box

0 7 . 3 Explain why the swing will eventually stop.

[2 marks]

0 7 . 4 **Figure 11** shows two children on the same swing. Both children oscillate at the same frequency.

Figure 11



Explain why child 1 will have a different maximum kinetic energy to child 2.

[2 marks]

13

Turn over ►

Do not write outside the box

0 8 The isotope caesium-137 (Cs) is radioactive.

A nucleus of caesium-137 emits a beta particle when it decays. This forms a nucleus of barium (Ba).

0 8 . 1 Complete the nuclear equation for the decay of caesium-137

[2 marks]

$$^{137}_{55}\mathrm{Cs} \longrightarrow {}^{\_\_\_}_{\_\_\_}\mathrm{Ba} + {}^{\_\_\_}_{\_\_\_}\beta$$

0 8 . 2 The half-life of caesium-137 is 30 years.

A sample of caesium-137 is 120 years old and has an activity of 24 Bq.

Calculate the initial activity of the sample of caesium-137

[3 marks]

____

____

____

____

____

Initial activity = ____________________ Bq

The beta radiation emitted by caesium-137 is used to kill bacteria on fruit.

Fruit remains fresh for longer because the fruit is irradiated.

0 8 . 3 The fruit is **not** contaminated by the caesium-137

Explain the difference between irradiation of an object and radioactive contamination of an object.

[2 marks]

0 8 . 4 Give **one** hazard and **one** precaution when handling caesium-137 for the irradiation of fruit.

[2 marks]

Hazard

Precaution

9

END OF QUESTIONS

Do not write outside the box

There are no questions printed on this page

DO NOT WRITE ON THIS PAGE
ANSWER IN THE SPACES PROVIDED

Do not write outside the box

Question number	**Additional page, if required. Write the question numbers in the left-hand margin.**

Do not write outside the box

Question number	**Additional page, if required. Write the question numbers in the left-hand margin.**